Finite-Cutoff Active Response Transport Theorem

1 Finite-Cutoff Active Response Transport Theorem

This section assembles a finite-cutoff transport theorem for the odd descendant / Hamiltonization lane of the 2T Main assumption package Definition 4.2. Accordingly, the phrase "accepted tick" remains literal UOM cutoff language throughout this note: the transport theorem is pointwise on the accepted branch at fixed Λ_k.

Remark 1.1 (Physical odd sector and branchwise ambient-space conventions). Throughout this note, the superscript $^-$ refers to physical Γ-odd parity on the characterization layer, not to cohomological oddness. Every ambiguity direction appearing below is therefore understood after passage to the physically odd characterization sector $E^-_{\Lambda_k}$.

All branchwise continuity statements are made after the standard accepted-branch identifications of the relevant finite-cutoff source spaces, covariance-tangent spaces, and physical subspaces with fixed ambient finite-cutoff spaces. Operator-norm and Fisher-norm continuity are always understood with respect to those identifications.

Definition 1.2 (Odd ambiguity subspace on the characterization layer)**.** For each accepted tick k, let

$$\mathcal{A}^-_{\Lambda_k} \subset E^-_{\Lambda_k}$$

denote the closed linear span of the odd characterization-layer projections of the descendant ambiguity directions

$$\big(Q_t X + Q_u Y\big)^-, \qquad \big(\partial_\mu K^\mu\big)^-, \qquad \Delta^-_{\mathrm{ct}},$$

where K^μ is boundary-admissible on the characterization layer and Δ^-_{ct} is admissible in the sense of **??**. Equivalently, $\mathcal{A}^-_{\Lambda_k}$ is the odd ambiguity space already placed inside $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by **??**.

Definition 1.3 (Dominant compressed response line and scalar coefficient)**.** Fix an accepted tick k. Let

$$u^{\mathrm{LoG}}_{\Lambda_k}$$

and

$$P^{\mathrm{LoG}}_{-,\Lambda_k}$$

be the canonically selected compressed odd line and projector furnished by **??**. Define the physical covariance tangent induced by that line by

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\, \mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

and, whenever this tangent is nonzero, write

$$L^{\mathrm{dom}}_{\Lambda_k} := \mathbb{R}\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$$

for its Fisher-normalized line. The raw dominant-response coefficient is

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F) := \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}, \qquad F \in E^-_{\Lambda_k}. \tag{1}$$

Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, equivalently whenever dominant-line realization holds at Λ_k, define the normalized dominant-response coefficient by

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) := \kappa^{-1/2}_{\Lambda_k}\,\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F), \qquad F \in E^-_{\Lambda_k}. \tag{2}$$

Proposition 1.4 (Well-posedness and continuity of the dominant-response coefficients)**.** *Fix an accepted tick k. Then the raw dominant-response coefficient*

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is continuous and linear. If dominant-line realization holds at Λ_k, then the normalized dominant-response coefficient

$$\rho^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is also continuous and linear.

Proof. By **??**, the intrinsic response map $\mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ is continuous and linear. Pairing its physical projection against the fixed covariance tangent $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ in the Fisher metric therefore gives a continuous linear functional, which is exactly $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$. If dominant-line realization holds, then $\kappa_{\Lambda_k} > 0$ by proposition 1.7, so $\rho^{\mathrm{dom}}_{\Lambda_k}$ is obtained from $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$ by multiplication by the fixed scalar $\kappa^{-1/2}_{\Lambda_k}$. □

Dominant-line realization

The compressed spectral-selection note already supplies the one-dimensional source line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. The present transport note needs only the nonvanishing and branchwise continuity of its *physical* covariance push-forward.

Definition 1.5 (Compressed-line transversality)**.** For an accepted tick k, say that the welded response is *transversely physical on the compressed line* if

$$\Pi_{\mathrm{phys},\Lambda_k}\,\mathsf{T}_{\Lambda_k}\restriction_{\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}} \neq 0. \tag{3}$$

Equivalently, the unique generator $u^{\mathrm{LoG}}_{\Lambda_k}$ of the compressed odd line has nonzero physical covariance response.

Definition 1.6 (Dominant-response norm)**.** Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is defined, set

$$\kappa_{\Lambda_k} := \langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{4}$$

This is the Fisher squared norm of the physical covariance tangent induced by the compressed welded LoG source line.

Proposition 1.7 (Equivalent formulations of dominant-line realization)**.** *For a fixed accepted tick k, the following are equivalent.*

1. *the compressed welded LoG line is transversely physical in the sense of definition 1.5;*

2. *the distinguished physical covariance tangent is nonzero,*

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0;$$

3. *the dominant-response norm is strictly positive,*

$$\kappa_{\Lambda_k} > 0;$$

4. *there exists a physical covariance tangent* $B_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$ *such that*

$$\langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, B_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} \neq 0;$$

5. *in any Gaussian completion realization of* **??**, *the completion-sector covariance tangent representing* $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ *is nonzero.*

Proof. The equivalence of (1) and (2) is immediate from (3) and the definition

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} = \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

because $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is the one-dimensional line generated by $u^{\mathrm{LoG}}_{\Lambda_k}$.

The equivalence of (2) and (3) follows from positive definiteness of the Fisher metric on the covariance tangent space: a vector has positive Fisher norm if and only if it is nonzero.

Statement (3) implies (4) by taking $B_{\Lambda_k} := \delta C^{\mathrm{LoG}}_{\Lambda_k}$, which belongs to the physical tangent space by construction. Conversely, if (4) holds then $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ cannot vanish, so (2) holds.

Finally, **????** identify the abstract covariance tangent with any Gaussian completion representative of the same tangent object. Therefore the abstract tangent is nonzero exactly when its completion realization is nonzero, giving the equivalence of (2) and (5). □

Theorem 1.8 (Branchwise realization and persistence of the dominant line)**.** *Let* I *be a connected accepted-branch segment in cutoff space. Assume that:*

1. *the compressed projector* $P^{\mathrm{LoG}}_{-,\Lambda}$ *varies continuously on* I*;*

2. *the physical welded response operator*

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_{\Lambda}$$

   is continuous on I *in the operator topology induced by the Fisher norm on covariance tangents;*

3. *there exists a constant* $c_I > 0$ *such that*

$$\kappa_{\Lambda} \geq c_I \qquad \forall \Lambda \in I.$$

Then:

1. $\delta C^{\mathrm{LoG}}_{\Lambda} \neq 0$ *for every* $\Lambda \in I$*;*

2. *the normalized dominant vector*

$$\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda} := \kappa_{\Lambda}^{-1/2}\, \delta C^{\mathrm{LoG}}_{\Lambda}$$

   is continuous on I*;*

3. *the dominant physical line*

$$L_\Lambda^{\mathrm{dom}} = \mathbb{R}\,\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$$

is canonically transported along I;

4. *the Fisher-orthogonal rank-one projector*

$$\Pi_\Lambda^{\mathrm{dom}} := \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \otimes \widehat{\delta C}_\Lambda^{\mathrm{LoG}}$$

depends continuously on $\Lambda \in I$.

Proof. By **??**, the compressed projector $P_{-,\Lambda}^{\mathrm{LoG}}$ and hence the normalized source vector u_Λ^{LoG} vary continuously on the branch. By the assumed continuity of $\Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda$, the map

$$\Lambda \longmapsto \delta C_\Lambda^{\mathrm{LoG}} = \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda[u_\Lambda^{\mathrm{LoG}}]$$

is continuous in the Fisher norm.

The lower bound $\kappa_\Lambda \geq c_I > 0$ implies $\delta C_\Lambda^{\mathrm{LoG}} \neq 0$ for every $\Lambda \in I$, proving item (1). Since the normalization map $v \mapsto v/\|v\|_{C_*}$ is continuous on the complement of the zero section, continuity of $\delta C_\Lambda^{\mathrm{LoG}}$ and the positive lower bound on its norm imply continuity of $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$, proving item (2). Items (3) and (4) are immediate consequences of item (2). □

Corollary 1.9 (Local persistence from one-tick transversality)**.** *Assume the continuity hypotheses of theorem 1.8. If the compressed line is transversely physical at one accepted tick Λ_{k_0}, then there exists a neighborhood I_{k_0} of Λ_{k_0} on the same accepted branch such that*

$$\delta C_\Lambda^{\mathrm{LoG}} \neq 0 \qquad \forall \Lambda \in I_{k_0}.$$

Hence the dominant physical line is locally well defined and continuously transported near Λ_{k_0}.

Proof. By proposition 1.7, transversality at Λ_{k_0} is equivalent to $\kappa_{\Lambda_{k_0}} > 0$. Continuity of κ_Λ then gives a neighborhood I_{k_0} and a constant $c_{k_0} > 0$ such that $\kappa_\Lambda \geq c_{k_0}$ for all $\Lambda \in I_{k_0}$. Apply theorem 1.8. □

Remark 1.10 (Practical witness criteria)*.* For applications, any one of the following suffices to discharge dominant-line realization at a fixed accepted tick:

1. a direct lower bound $\kappa_{\Lambda_k} > 0$;
2. a physical witness tangent $B_{\Lambda_k} \in \mathcal{T}_{\Lambda_k}^{\mathrm{phys}}$ with nonzero Fisher pairing against $\delta C_{\Lambda_k}^{\mathrm{LoG}}$;
3. a Gaussian completion calculation showing that the completion covariance of the realized active odd mode changes nontrivially under the compressed welded LoG source.

Each witness proves the same theorem-level fact by proposition 1.7.

Descent to the ambiguity quotient

Once the dominant line has been realized, the next task is purely structural: show that the dominant-response coefficient is insensitive to ambiguity moves. This is the exact point at which the transported descendant class becomes an affine class in the sense required by Main rather than a collection of representatives.

Definition 1.11 (Dominant-response quotient and descended descendant point)**.** For each accepted tick k, define the dominant-response quotient by

$$\mathfrak{D}^-_{\Lambda_k} := E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k},$$

with quotient map

$$q_{\Lambda_k} : E^-_{\Lambda_k} \to \mathfrak{D}^-_{\Lambda_k}.$$

If

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k},$$

then its image in the quotient is a single point,

$$[\![\Xi^{(1)}_{\Lambda_k}]\!] := q_{\Lambda_k}\big(J_{0,\Lambda_k}\big) \in \mathfrak{D}^-_{\Lambda_k},$$

independent of the chosen representative J_{0,Λ_k}.

Proposition 1.12 (Universal descent criterion for the dominant-response coefficient)**.** *Assume dominant-line realization at an accepted tick k, so that $\rho^{\mathrm{dom}}_{\Lambda_k}$ is defined by (2). Then the following are equivalent.*

1. *there exists a unique continuous linear functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

   such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k};$$

2. *the ambiguity subspace is contained in the kernel:*

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k};$$

3. *the dominant-response coefficient is constant on each ambiguity coset:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F + A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F) \qquad \forall F \in E^-_{\Lambda_k},\ \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Proof. The equivalence of (2) and (3) is immediate from linearity:

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F + A) - \rho^{\mathrm{dom}}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(A).$$

If (2) holds, define

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(F)\big) := \rho^{\mathrm{dom}}_{\Lambda_k}(F).$$

This is well defined precisely because changing representatives by an element of $\mathcal{A}^-_{\Lambda_k}$ does not change the value of $\rho^{\mathrm{dom}}_{\Lambda_k}$. Linearity and continuity are inherited from $\rho^{\mathrm{dom}}_{\Lambda_k}$, and uniqueness follows because q_{Λ_k} is surjective.

Conversely, if (1) holds and $A \in \mathcal{A}^-_{\Lambda_k}$, then $q_{\Lambda_k}(A) = 0$, hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(A)\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(0) = 0.$$

Thus (1) implies (2). □

Corollary 1.13 (Constancy on the transported descendant affine class)**.** *Assume proposition 1.12. Then*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

In particular, once the descended quotient functional exists, the descendant affine class carries a single unambiguous dominant-response value.

Proof. Every $J \in [\Xi^{(1)}_{\Lambda_k}]$ has the form $J = J_{0,\Lambda_k} + A$ with $A \in \mathcal{A}^-_{\Lambda_k}$, so

$$q_{\Lambda_k}(J) = q_{\Lambda_k}(J_{0,\Lambda_k}) = [\![\Xi^{(1)}_{\Lambda_k}]\!].$$

Apply the factorization identity of proposition 1.12. □

Remark 1.14 (False stronger form and normalized special case)*.* The quotient statement above is intrinsically a statement about the descended dominant scalar

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big),$$

not about a bare χ-compatibility condition. At the weak transported-interface layer one has

$$\chi_{\Lambda_k} = \nu^{-1}_{\Lambda_k}\rho^{\mathrm{dom}}_{\Lambda_k},$$

so a representative-search condition of the form

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = s$$

is equivalent to

$$\chi_{\Lambda_k}(J) = \nu^{-1}_{\Lambda_k} s,$$

not in general to $\chi_{\Lambda_k}(J) = s$. Only after an explicit normalization, for example $\nu_{\Lambda_k} = 1$, may the same scalar comparison be rewritten in descended $\bar{\chi}_{\Lambda_k}$ form without changing the value. The finite-cutoff Wolfram audit contains exact countermodels to the stronger unscaled χ-level claim.

Proposition 1.15 (Generator-by-generator reduction)**.** *Assume dominant-line realization at an accepted tick k. Suppose the dominant-response coefficient vanishes on the three generating ambiguity families:*

1. *for every odd-projected BRST/gauge-exact direction $\big(Q_t X + Q_u Y\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(Q_t X + Q_u Y\big)^-\big) = 0;$$

2. *for every odd-projected boundary-admissible divergence $\big(\partial_\mu K^\mu\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(\partial_\mu K^\mu\big)^-\big) = 0;$$

3. *for every admissible odd counterterm Δ^-_{ct},*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Then

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k},$$

hence $\rho^{\mathrm{dom}}_{\Lambda_k}$ descends uniquely to the quotient $\mathfrak{D}^-_{\Lambda_k}$.

Proof. By definition 1.2, $\mathcal{A}^-_{\Lambda_k}$ is the closed linear span of the three families listed in the statement. Since $\rho^{\mathrm{dom}}_{\Lambda_k}$ is linear, it vanishes on the algebraic span of those generators. Since it is also continuous, it vanishes on the closure of that span. Therefore every element of $\mathcal{A}^-_{\Lambda_k}$ lies in the kernel. Now apply proposition 1.12. □

Theorem 1.16 (Descent to the ambiguity quotient)**.** *Fix an accepted tick k and assume dominant-line realization. If the three vanishing statements of proposition 1.15 hold, then there exists a unique descended dominant functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k},$$

and the transported descendant affine class has a single well-defined quotient value

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big).$$

Proof. Apply proposition 1.15 and corollary 1.13. □

Nonzero overlap on the descendant class

After dominant-line realization and quotient descent, the descendant class defines a single point in the ambiguity quotient, and the dominant-response functional descends to that quotient. This subsection studies whether that descended descendant point is *active*, i.e. not annihilated by the descended dominant functional.

Definition 1.17 (Dominant covariance projector)**.** Assume dominant-line realization at an accepted tick k. Define the Fisher-orthogonal rank-one projector onto the dominant physical line by

$$\Pi^{\mathrm{dom}}_{\Lambda_k}(X) := \big\langle\!\big\langle X,\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}, \qquad X \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}. \tag{5}$$

Proposition 1.18 (Representative-level witness criteria)**.** *Assume dominant-line realization at an accepted tick k. For any odd generator $F \in E^-_{\Lambda_k}$, the following are equivalent.*

1. *the dominant-response coefficient is nonzero:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) \neq 0;$$

2. *the dominant projection of the physical covariance response is nonzero:*

$$\Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) \neq 0;$$

3. *the physical covariance response has nonzero Fisher pairing with the dominant tangent:*

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \neq 0;$$

4. *in any Gaussian completion realization of* **??**, *the completion representative of $\Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F)$ has nonzero component on the realized active odd line.*

Proof. By definition 1.17,

$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\,\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}.$$

Since $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, this proves the equivalence of (1) and (2).

Statement (3) is equivalent to (1) because

$$\left\langle\!\left\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\right\rangle\!\right\rangle_{C_{*,\Lambda_k}} = \|\delta C^{\mathrm{LoG}}_{\Lambda_k}\|_{C_{*,\Lambda_k}}\,\rho^{\mathrm{dom}}_{\Lambda_k}(F),$$

and the prefactor is strictly positive by dominant-line realization.

Finally, by **????**, the abstract covariance tangent and its Gaussian completion representative describe the same physical response object in different coordinates. Therefore the abstract dominant projection is nonzero if and only if the completion representative has nonzero component on the realized active line, proving the equivalence of (2) and (4). □

Definition 1.19 (Descendant overlap margin)**.** Assume dominant-line realization and quotient descent at an accepted tick k. Define the descendant overlap margin by

$$\nu_{\Lambda_k} := \left|\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)\right|. \tag{6}$$

We say that the transported descendant class is *dominant-active* if $\nu_{\Lambda_k} > 0$.

Proposition 1.20 (Equivalent formulations of nonzero overlap on the descendant class)**.** *Assume dominant-line realization and quotient descent at an accepted tick k. Then the following are equivalent.*

1. *the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;*
2. *there exists one representative $J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ such that*
$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0;$$
3. *every representative $J \in [\Xi^{(1)}_{\Lambda_k}]$ has the same nonzero dominant-response value:*
$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \neq 0;$$
4. *the descended descendant point is not annihilated by the descended dominant functional:*
$$[\![\Xi^{(1)}_{\Lambda_k}]\!] \notin \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k};$$
5. *the descendant overlap margin is strictly positive:*
$$\nu_{\Lambda_k} > 0.$$

Proof. The equivalence of (1) and (2) follows immediately from proposition 1.18.

If (2) holds, then $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0$ by proposition 1.18. By corollary 1.13, every other representative of $[\Xi^{(1)}_{\Lambda_k}]$ has the same value, proving (3). Conversely, (3) obviously implies (1).

The equivalence of (3) and (4) is immediate from the identity

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}]$$

of corollary 1.13.

Finally, (4) and (5) are equivalent by the definition of ν_{Λ_k} in definition 1.19. □

Theorem 1.21 (Nonzero overlap on the descendant class)**.** *Fix an accepted tick k, and assume dominant-line realization together with quotient descent. If any one of the equivalent conditions in proposition 1.20 holds, then the following conclusions obtain. In particular, it is enough to construct a single representative*

$$J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

whose dominant projected covariance response is nonzero.

When this holds, the quotient descendant point is dominant-active, the scalar value

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)$$

is nonzero, and the sign

$$\varepsilon_{\Lambda_k} := \mathrm{sgn}\Big(\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)\Big) \in \{\pm 1\} \tag{7}$$

is well defined. Thus the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line.

Proof. The equivalence package is exactly proposition 1.20. Once one representative has nonzero dominant projected response, the same proposition gives the nonzero scalar value on the descended descendant point. The sign in (7) is therefore well defined. □

Conditional mod-null reentry on the selected scalar line

Definition 1.22 (Parity-stable mod-null subspace transverse to the selected scalar line)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A *parity-stable mod-null datum transverse to* $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ consists of a linear subspace

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

such that:

(a) $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;

(b) the selected scalar line is transverse to $N^{\mathrm{mn}}_{\Lambda_k}$:

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

Define

$$\mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k} := \mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$$

with quotient map

$$\pi^{\mathrm{mn}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Because $\mathfrak{D}^-_{\Lambda_k}$ is itself a quotient of the physical Γ-odd sector, the descended parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$. Thus the parity-stability clause yields a descended involution

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\pi^{\mathrm{mn}}_{\Lambda_k}(x)\big) := \pi^{\mathrm{mn}}_{\Lambda_k}(-x), \qquad x \in \mathfrak{D}^-_{\Lambda_k}.$$

Definition 1.23 (Dominant-null parity-stable mod-null datum)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A parity-stable mod-null datum

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is called *dominant-null* if it is contained in the kernel of the descended dominant functional:

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}.$$

It is understood that $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$, and that the associated quotient $\mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$ carries the descended involution from definition 1.22. The canonical dominant-kernel choice is isolated separately in definition 1.24. The transversality lemma below uses only this kernel-subordination property.

Definition 1.24 (Canonical dominant-null datum)**.** Fix an accepted tick k and assume dominant-line realization together with quotient descent. Define the *canonical dominant-null datum* by

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

This datum depends only on the descended dominant functional on the ambiguity quotient. No additional reduced-welded or completion-side choice is built into the definition.

Lemma 1.25 (Dominant-kernel criterion for selected-line transversality)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) \neq 0.$$

If

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is a dominant-null parity-stable mod-null datum in the sense of definition 1.23, then

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

So every dominant-null parity-stable mod-null datum is automatically transverse to the selected scalar line.

Proof. Let $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k}$. Because $\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$, there exists $a \in \mathbb{R}$ with $x = a\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$. Since $N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}$, we have

$$0 = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big).$$

The coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, so $a = 0$. Therefore $x = 0$, proving $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$. $\square$

Theorem 1.26 (The dominant kernel is the canonical dominant-null datum)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}) \neq 0.$$

Define

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

Then:

1. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;*
2. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is a dominant-null parity-stable mod-null datum in the sense of definition 1.23;*
3. *the selected scalar line is automatically transverse to the canonical dominant-null datum:*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{dom-null}}_{\Lambda_k} = \{0\}.$$

Hence the canonical dominant kernel is admissible as the mod-null datum for the conditional reentry theorem.

Proof. Let $x \in N^{\mathrm{dom-null}}_{\Lambda_k}$. Because the descended physical parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$, we have

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(-x) = -\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = 0.$$

So $-x \in N^{\mathrm{dom-null}}_{\Lambda_k}$, proving parity stability. Item (2) is then exactly definition 1.23 with $N^{\mathrm{mn}}_{\Lambda_k} = N^{\mathrm{dom-null}}_{\Lambda_k}$. Item (3) is lemma 1.25 applied to this canonical choice. □

Lemma 1.27 (Transversality reduces mod-null reentry to quotient construction)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}) \neq 0.$$

Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ in the sense of definition 1.22. Define

$$\ell^{\mathrm{sel,mn}}_{\Lambda_k} := \pi^{\mathrm{mn}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_k} := \mathbb{R}\,\ell^{\mathrm{sel,mn}}_{\Lambda_k} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Then:

1. *the restriction of $\pi^{\mathrm{mn}}_{\Lambda_k}$ to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ is injective;*

2. $L^{\mathrm{sel,mn}}_{\Lambda_k}$ *is a well-defined one-dimensional line in the mod-null quotient, generated by the nonzero class* $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$*;*

3. *the mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k} : L^{\mathrm{sel,mn}}_{\Lambda_k} \longrightarrow \mathbb{R}\, u^{\mathrm{LoG}}_{\Lambda_k}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}\big(a\, \ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) := a\, \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)\, u^{\mathrm{LoG}}_{\Lambda_k}$$

is well defined and faithful;

4. *the descended involution acts by sign:*

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k}.$$

In particular, once the transversality condition $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$ *is known, the remaining mod-null reentry step reduces to quotient construction plus parity descent.*

Proof. If $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k}$ and $\pi^{\mathrm{mn}}_{\Lambda_k}(x) = 0$, then $x \in N^{\mathrm{mn}}_{\Lambda_k}$ by definition of the quotient kernel. Hence $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$, which proves injectivity. Since $\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \neq 0$, its image $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$ is therefore nonzero, and the image of a one-dimensional line under an injective linear map is again one-dimensional. This proves items (1) and (2).

Injectivity on $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ implies that every element of $L^{\mathrm{sel,mn}}_{\Lambda_k}$ has a unique expression $a\, \ell^{\mathrm{sel,mn}}_{\Lambda_k}$, so the displayed formula defines $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}$ unambiguously. Its coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, and $u^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ by **??**. Therefore $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}(a\, \ell^{\mathrm{sel,mn}}_{\Lambda_k}) = 0$ implies $a = 0$, so the map is faithful. This proves item (3).

By construction of the descended involution,

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) = \Gamma^{\mathrm{mn}}_{\Lambda_k}\Big(\pi^{\mathrm{mn}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)\Big) = \pi^{\mathrm{mn}}_{\Lambda_k}\big(-\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k},$$

which proves item (4). □

Theorem 1.28 (Conditional mod-null reentry on the selected scalar line)**.** *Fix an accepted tick* k *and the next accepted receiver step* $k+1$*. Assume the hypotheses and notation of* **??***. In particular, let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} = \mathbb{R}\, \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}$$

be the nonzero selected scalar descendant line supplied there. Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}}$ *in the sense of definition 1.22. Define*

$$\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \pi^{\mathrm{mn}}_{\Lambda_{k+1}}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}\big)$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \mathbb{R}\, \ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_{k+1}}.$$

Then:

1. $L^{\mathrm{sel,mn}}_{\Lambda_{k+1}}$ *is a well-defined one-dimensional selected descendant line in the mod-null quotient;*

2. *the induced mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}} : L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \longrightarrow \mathbb{R}\, u^{\mathrm{LoG}}_{\Lambda_{k+1}}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}}\big(a\, \ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}\big) := a\, \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}\big)\, u^{\mathrm{LoG}}_{\Lambda_{k+1}}$$

is well defined and faithful;

3. the descended involution acts by sign:

$$\Gamma^{\mathrm{mn}}_{\Lambda_{k+1}}(\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}) = -\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}.$$

Proof. By **??**,

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0 \qquad \text{and} \qquad \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So every hypothesis of lemma 1.27 is in force at Λ_{k+1}. Applying that lemma gives items (1)–(3) immediately. □

Corollary 1.29 (Canonical dominant-kernel discharge of the mod-null reentry step)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses and notation of* **??**. *Define the canonical dominant-null datum*

$$N^{\mathrm{dom-null}}_{\Lambda_{k+1}} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

Then $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is admissible as the mod-null datum of theorem 1.28, the transversality hypothesis of that theorem is automatic, and hence the conclusions of that theorem hold with this canonical datum.

Proof. By **??**,

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So theorem 1.26 applies at Λ_{k+1}. It shows that $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is a dominant-null parity-stable mod-null datum and that

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} \cap N^{\mathrm{dom-null}}_{\Lambda_{k+1}} = \{0\}.$$

This is exactly the admissibility and transversality clause required by theorem 1.28, so that theorem now applies directly with the canonical datum. □

Remark 1.30 (Conditional character of the reentry step)*.* theorem 1.28 is a quotient-level reentry statement. After lemma 1.25, theorem 1.26, and corollary 1.29, the canonical dominant-null datum is fixed as $\ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}$ on $\mathfrak{D}^{-}_{\Lambda_{k+1}}$. What remains is passage to the mod-null quotient together with parity descent on the odd line. Accordingly, the theorem should not be read as a statement on the raw descendant ambiguity quotient or on any stronger physical quotient.

Branchwise orientation transport

This subsection studies transport of the descended descendant sign along a connected accepted branch. The sign can be transported once the descended descendant value varies continuously with the cutoff. The compressed spectral-selection theorem already gives continuity of the dominant projector, and theorem 1.8 gives continuity of the dominant physical line. We combine those inputs with continuity of the projected descendant response.

Definition 1.31 (Branchwise descendant overlap function)**.** Let I be a connected accepted-branch segment. Assume dominant-line realization and quotient descent for every $\Lambda \in I$. Suppose there is a branchwise choice of representatives

$$J^{\mathrm{desc}}_{\Lambda} \in [\Xi^{(1)}_{\Lambda}] \qquad (\Lambda \in I)$$

such that the projected covariance response

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_{\Lambda}(J^{\mathrm{desc}}_{\Lambda})$$

is continuous in the Fisher topology. Define the associated branchwise descendant overlap function by

$$m_I(\Lambda) := \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big). \tag{8}$$

Proposition 1.32 (Continuity of the descended overlap scalar)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of theorem 1.8 on I;*
2. *quotient descent holds for every $\Lambda \in I$;*
3. *the branchwise choice of representatives in definition 1.31 exists.*

Then the scalar function $m_I : I \to \mathbb{R}$ is well defined, independent of the chosen branchwise representatives, and continuous. Moreover,

$$|m_I(\Lambda)| = \nu_\Lambda \qquad \forall \Lambda \in I. \tag{9}$$

Proof. Independence of the chosen branchwise representatives is exactly the coset constancy proved in corollary 1.13: if $J_\Lambda'^{\mathrm{desc}} - J_\Lambda^{\mathrm{desc}} \in \mathcal{A}_\Lambda^-$, then

$$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda'^{\mathrm{desc}}) = \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big).$$

Thus m_I is well defined.

By theorem 1.8, the normalized dominant vector $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$ and the dominant projector $\Pi_\Lambda^{\mathrm{dom}}$ depend continuously on $\Lambda \in I$. By assumption, the response vector

$$X_\Lambda := \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_\Lambda(J_\Lambda^{\mathrm{desc}})$$

is also continuous on I. Using definition 1.17, we have

$$\Pi_\Lambda^{\mathrm{dom}} X_\Lambda = m_I(\Lambda)\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}}.$$

The left-hand side is continuous because both factors are continuous, while $\widehat{\delta C}_\Lambda^{\mathrm{LoG}} \neq 0$ for all $\Lambda \in I$ by theorem 1.8. Therefore the scalar coefficient $m_I(\Lambda)$ is continuous.

Finally, (9) is just the definition of ν_Λ from definition 1.19 together with (8). □

Theorem 1.33 (Branchwise orientation transport on the descendant class)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 1.32. Then:*

1. *on every connected subsegment $I' \subset I$ such that*

$$\nu_\Lambda > 0 \qquad \forall \Lambda \in I',$$

the sign

$$\varepsilon_\Lambda := \operatorname{sgn} m_I(\Lambda) = \operatorname{sgn}\Big(\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\Big) \in \{\pm 1\} \tag{10}$$

is constant on I';

2. *if $\nu_{\Lambda_0} > 0$ at one point $\Lambda_0 \in I$, then there exists a neighborhood $I_0 \subset I$ of Λ_0 on which ε_Λ is constant and equal to ε_{Λ_0};*

3. *under the present hypotheses, a change of sign can occur only at a cutoff where the descendant overlap closes* $\nu_\Lambda = 0$.

In particular, once dominant-line realization, quotient descent, and nonzero overlap are established on an accepted branch segment and the projected descendant response varies continuously there, the branchwise sign statement of theorem 1.56 follows.

Proof. By proposition 1.32, $m_I(\Lambda)$ is a continuous real-valued function on I. On any connected subsegment I' where $\nu_\Lambda = |m_I(\Lambda)| > 0$, the function m_I never vanishes. A continuous nonvanishing real-valued function on a connected set has constant sign, proving item (1).

Item (2) is the local version of the same argument: if $m_I(\Lambda_0) \neq 0$, continuity gives a neighborhood I_0 on which m_I keeps the sign it has at Λ_0.

For item (3), any sign change would force m_I to pass through zero by the intermediate value theorem, hence $\nu_\Lambda = |m_I(\Lambda)| = 0$ at some intermediate cutoff. □

Corollary 1.34 (Local orientation persistence from one-tick activity)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 1.32. If the transported descendant class is dominant-active at one cutoff $\Lambda_0 \in I$, i.e.*

$$\nu_{\Lambda_0} > 0,$$

then there exists a neighborhood $I_0 \subset I$ of Λ_0 such that

$$\varepsilon_\Lambda = \varepsilon_{\Lambda_0} \qquad \forall \Lambda \in I_0.$$

Proof. This is exactly item (2) of theorem 1.33. □

Witness-indexed seam realization and scalar bridge

The bridge used below factors through a common LS witness datum rather than a direct identification of a UOM seam source with a descendant representative. On the UOM side one realizes a filler tag together with a detector witness by an admissible odd welded source, measures the resulting dominant physical pairing, and then compares that scalar with the odd representative attached to the same witness.

Definition 1.35 (Accepted record payload of a welded odd source)**.** Fix an accepted tick k. For an admissible welded odd trace-channel source τ, let

$$\rho_{\Lambda_k}[\tau]$$

denote the receiver state at cutoff Λ_k produced by that source in the welded UOM channel. Define its accepted record payload by

$$\mathcal{P}^{\mathrm{acc}}_{\Lambda_k}[\tau] := \Pi_{\Delta(\Lambda_k)}\big(\rho_{\Lambda_k}[\tau]\big). \tag{11}$$

This is the diagonal/record part retained by the finite-cutoff acceptance map.

Definition 1.36 (Witness-indexed seam realization at an accepted tick)**.** Fix an accepted tick k. Let y be a witness on the LS difference object, and write $\mathrm{flip}(y)$ for the induced LS involution. A *witness-indexed seam realization* at Λ_k is a choice of admissible welded odd trace-channel sources

$$\tau^{f,y}_{\Lambda_k}, \qquad f \in \{\mathrm{diag}, \mathrm{bridge}\},$$

such that:

1. **filler toggle changes the odd seam sign:**

$$\tau_{\Lambda_k}^{\mathrm{bridge},y} = -\,\tau_{\Lambda_k}^{\mathrm{diag},y};$$

2. **witness flip changes the odd seam sign:**

$$\tau_{\Lambda_k}^{f,\mathrm{flip}(y)} = -\,\tau_{\Lambda_k}^{f,y} \qquad \forall f \in \{\mathrm{diag}, \mathrm{bridge}\};$$

3. **accepted record payload is unchanged under filler toggle:**

$$\mathcal{P}_{\Lambda_k}^{\mathrm{acc}}\big[\tau_{\Lambda_k}^{\mathrm{bridge},y}\big] = \mathcal{P}_{\Lambda_k}^{\mathrm{acc}}\big[\tau_{\Lambda_k}^{\mathrm{diag},y}\big].$$

Remark 1.37 (Accepted-record invariance versus seam-scalar use). The accepted-record invariance clause in definition 1.36 encodes the physical indistinguishability of filler toggle at the accepted-record layer. The seam-scalar theorems below use only the induced odd sign relations in the source channel.

Definition 1.38 (Raw and normalized seam pairings)**.** Fix an accepted tick k and a witness-indexed seam realization as in definition 1.36. For $f \in \{\mathrm{diag}, \mathrm{bridge}\}$ and witness y, define the *raw seam pairing* by

$$\widetilde{s}_{\Lambda_k}(f,y) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{f,y}\big],\, \delta C_{\Lambda_k}^{\mathrm{LoG}} \big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{12}$$

Whenever dominant-line realization holds so that $\widehat{\delta C}_{\Lambda_k}^{\mathrm{LoG}}$ is defined, also set

$$s_{\Lambda_k}(f,y) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{f,y}\big],\, \widehat{\delta C}_{\Lambda_k}^{\mathrm{LoG}} \big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{13}$$

Proposition 1.39 (Filler and witness parity of the seam scalar)**.** *Fix an accepted tick k and assume a witness-indexed seam realization at Λ_k. Then the raw seam pairing satisfies*

$$\widetilde{s}_{\Lambda_k}(\mathrm{bridge}, y) = -\,\widetilde{s}_{\Lambda_k}(\mathrm{diag}, y), \qquad \widetilde{s}_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,\widetilde{s}_{\Lambda_k}(f, y). \tag{14}$$

If dominant-line realization also holds, then the same identities hold for the normalized seam pairing:

$$s_{\Lambda_k}(\mathrm{bridge}, y) = -\,s_{\Lambda_k}(\mathrm{diag}, y), \qquad s_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,s_{\Lambda_k}(f, y). \tag{15}$$

Proof. The map $\tau \mapsto \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau]$ is linear because T_{Λ_k} and $\Pi_{\mathrm{phys},\Lambda_k}$ are linear. The Fisher pairing is bilinear. Applying the sign relations of definition 1.36 therefore gives

$$\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{\mathrm{bridge},y}\big] = -\,\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{\mathrm{diag},y}\big]$$

and

$$\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{f,\mathrm{flip}(y)}\big] = -\,\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{f,y}\big].$$

Pairing with $\delta C_{\Lambda_k}^{\mathrm{LoG}}$ proves (14). If dominant-line realization holds, then $\widehat{\delta C}_{\Lambda_k}^{\mathrm{LoG}}$ is a fixed nonzero normalization of the same vector, so the same argument proves (15). □

Definition 1.40 (Detector-factorized seam coupling)**.** Fix an accepted tick k. A witness-indexed seam realization is called *detector-factorized* if there exists a single admissible welded odd trace-channel source

$$\tau_{\Lambda_k}^{\mathrm{unit}}$$

and the LS detector factor

$$\overline{\mathrm{obs}} : \text{LS difference object} \to \mathbb{R}$$

such that, for every filler tag f and witness y,

$$\tau_{\Lambda_k}^{f,y} = \sigma_{\mathrm{fill}}(f)\,\overline{\mathrm{obs}}(y)\,\tau_{\Lambda_k}^{\mathrm{unit}}, \qquad \sigma_{\mathrm{fill}}(\mathrm{diag}) = +1, \quad \sigma_{\mathrm{fill}}(\mathrm{bridge}) = -1. \tag{16}$$

Theorem 1.41 (Seam faithfulness scalar)**.** *Fix an accepted tick k and assume a detector-factorized witness-indexed seam realization at Λ_k. Define the fixed-cutoff seam coefficient by*

$$\widetilde{c}_{\Lambda_k} := \left\langle\!\!\left\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \right\rangle\!\!\right\rangle_{C_{*,\Lambda_k}}. \tag{17}$$

Then, for every filler tag f and witness y,

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\, \widetilde{c}_{\Lambda_k}\, \overline{\mathrm{obs}}(y). \tag{18}$$

In particular, if

$$\widetilde{c}_{\Lambda_k} \neq 0 \qquad \text{and} \qquad \overline{\mathrm{obs}}(y) \neq 0,$$

then the raw seam pairing is nonzero:

$$\widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

Consequently the compressed welded LoG line is transversely physical at Λ_k, hence dominant-line realization holds at that cutoff by proposition 1.7. When dominant-line realization holds, the normalized seam coefficient

$$c_{\Lambda_k} := \kappa_{\Lambda_k}^{-1/2}\, \widetilde{c}_{\Lambda_k} \tag{19}$$

is defined and

$$s_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\, c_{\Lambda_k}\, \overline{\mathrm{obs}}(y). \tag{20}$$

Proof. Substitute the detector-factorized form (16) into the definition (12). Linearity of T_{Λ_k} and bilinearity of the Fisher pairing give

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\, \overline{\mathrm{obs}}(y)\, \left\langle\!\!\left\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \right\rangle\!\!\right\rangle_{C_{*,\Lambda_k}},$$

which is exactly (18).

If $\widetilde{c}_{\Lambda_k} \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$, then $\widetilde{s}_{\Lambda_k}(f,y) \neq 0$. Set

$$B_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}\big[\tau^{f,y}_{\Lambda_k}\big].$$

Then

$$\left\langle\!\!\left\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, B_{\Lambda_k} \right\rangle\!\!\right\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

By item (4) of proposition 1.7, this proves transversality of the compressed line and hence dominant-line realization at Λ_k.

Once dominant-line realization holds, divide (18) by $\|\delta C^{\mathrm{LoG}}_{\Lambda_k}\|_{C_{*,\Lambda_k}} = \kappa_{\Lambda_k}^{1/2}$ to obtain (20). □

Remark 1.42 (Downstream role of the seam-coefficient hypothesis)*.* The fixed-cutoff nonvanishing hypothesis $\widetilde{c}_{\Lambda_k} \neq 0$ is exactly the input that, once dominant-line realization defines c_{Λ_k}, is imported downstream as the nondegeneracy clause $c_{\Lambda_k} \neq 0$ inside the delayed selected-line package. This note does not derive that clause from split-threshold analysis alone.

Definition 1.43 (Odd source representability on the characterization layer)**.** Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source. We say that τ is *representable on the characterization layer* if there exists an odd sufficiently local generator

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}$$

such that its intrinsic physical covariance response agrees with the UOM welded response:

$$\Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau]. \tag{21}$$

This is the strong source-insertion realization statement.

Theorem 1.44 (UOM source-insertion representability theorem)**.** *Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source at cutoff Λ_k, in the finite-cutoff UOM sense: its receiver-head profile is band-limited at Λ_k and compactly supported inside the accepted slab. Then there exists a canonically associated odd sufficiently local generator*

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}$$

such that

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau]. \tag{22}$$

In particular, every witness-indexed seam source used in the present note is representable on the characterization layer.

Proof. By the finite-cutoff UOM Peierls package, admissible receiver-head sources at cutoff Λ_k are band-limited boundary test data and the sufficiently local class $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}$ contains all compact smearings of local boundary densities. Let

$$F_{\tau,\Lambda_k}$$

be the linear source functional obtained by pairing the odd trace-channel source profile τ with the corresponding odd trace-channel boundary field on the receiver head. Because this is the linear case of the sufficiently local form and τ is physically odd,

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}.$$

Now apply the UOM source insertion map from the finite-cutoff retarded-kernel package. The source τ is sent to the bulk forcing term in the linearized equation; the unique retarded solution is obtained by the retarded Green operator; and boundary restriction gives the retarded covariance variation. By the UOM identification of the induced retarded kernel with the SK mixed second derivative, this covariance variation is exactly the welded source-to-covariance response

$$\mathsf{T}_{\Lambda_k}[\tau].$$

On the other hand, by the definition of the intrinsic covariance-response map, the matrix elements of $\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k})$ are computed by the Peierls action of F_{τ,Λ_k} on the sufficiently local quadratic probes representing boundary covariance entries. That Peierls variation is precisely the same retarded second-moment variation generated by the source insertion τ. Therefore

$$\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \mathsf{T}_{\Lambda_k}[\tau]$$

as covariance tangents. Applying the physical projector $\Pi_{\mathrm{phys},\Lambda_k}$ gives (22).

The witness-indexed seam sources of definition 1.36 are, by construction, live admissible welded odd trace-channel sources at the accepted tick, so the same argument applies to each of them. □

Proposition 1.45 (Dominant pairing for a representable odd source)**.** *Fix an accepted tick k and let τ be an admissible welded odd trace-channel source that is representable in the sense of theorem 1.44. Then*

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{23}$$

If dominant-line realization also holds, then

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau,\Lambda_k}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{24}$$

Proof. Equation (21) identifies the two physical covariance tangents. Pairing both sides with $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ gives (23). If dominant-line realization holds, pair with the normalized vector $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ instead and use the definition (2) of $\rho^{\mathrm{dom}}_{\Lambda_k}$ to obtain (24). □

Corollary 1.46 (Unit welded odd source representability and scalar identification)**.** *Fix an accepted tick k. Assume a detector-factorized witness-indexed seam realization at Λ_k in the sense of definition 1.40, and assume dominant-line realization at Λ_k. Then the unit welded odd source $\tau^{\mathrm{unit}}_{\Lambda_k}$ is representable on the characterization layer, with canonical representable generator*

$$F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k} \in E^-_{\Lambda_k},$$

and

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = c_{\Lambda_k}. \tag{25}$$

Proof. Representability of $\tau^{\mathrm{unit}}_{\Lambda_k}$ is the specialization of theorem 1.44 to the admissible welded odd trace source singled out by definition 1.40. Because dominant-line realization holds, proposition 1.45 applies to that representable source and gives the first equality in (25). The second equality is exactly the definition of the normalized seam coefficient c_{Λ_k} from theorem 1.41. □

Definition 1.47 (Scalar comparison shadow of the witness-indexed bridge)**.** Fix an accepted tick k. Let y be an LS witness and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source for a filler tag f. A *scalar comparison shadow* for that data is a choice of representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

such that the dominant raw pairing of the 2T-side representative agrees with the dominant raw pairing of the corresponding representable UOM source:

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{26}$$

When dominant-line realization holds, this is equivalently the equality of normalized dominant pairings

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big). \tag{27}$$

Definition 1.48 (Witness-indexed source-compatible descendant realization)**.** Fix an accepted tick k, a filler tag f, a witness y, and a witness-indexed seam source $\tau^{f,y}_{\Lambda_k}$. A representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

is called a *witness-indexed source-compatible descendant realization* if it realizes, on the same LS witness datum, the Limited Scope output and odd Hamiltonization clauses from item **(M1)** and item **(M4)** of Definition 4.2 and differs from the source-insertion representative only by an ambiguity direction:

$$J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} \in \mathcal{A}^-_{\Lambda_k}, \tag{28}$$

where $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ is the representable source generator furnished by theorem 1.44. Equivalently, the witness-indexed transported descendant class and the witness-indexed UOM source-insertion realization define the same point of the ambiguity quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$.

Theorem 1.49 (Same-witness descended-scalar comparison from a source-compatible descendant realization)**.** *Fix an accepted tick k, assume dominant-line realization at Λ_k, and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source. Assume:*

1. *the source $\tau^{f,y}_{\Lambda_k}$ is represented by the canonical generator $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ from theorem 1.44;*
2. *there exists a witness-indexed source-compatible descendant realization $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ in the sense of definition 1.48;*
3. *the three vanishing statements of proposition 1.15 hold at Λ_k.*

Then the raw and normalized comparison identities of definition 1.47 hold:

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{f,y}_{\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}, \tag{29}$$

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}). \tag{30}$$

Moreover,

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{31}$$

If quotient descent also holds at Λ_k, then the common value equals the descended dominant scalar on the transported descendant affine class:

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}([\![\Xi^{(1)}_{\Lambda_k}]\!]) = \rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}). \tag{32}$$

Proof. Set

$$A^{f,y}_{\Lambda_k} := J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}.$$

By source compatibility, $A^{f,y}_{\Lambda_k} \in \mathcal{A}^-_{\Lambda_k}$.

Because dominant-line realization holds and the three vanishing statements of proposition 1.15 are assumed, that proposition gives

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k}.$$

Hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} + A^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}),$$

proving (30).

Since dominant-line realization implies $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ and $\rho^{\mathrm{dom}}_{\Lambda_k} = \kappa^{-1/2}_{\Lambda_k}\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$, the raw pairing equality (29) is just the same identity multiplied by the positive factor $\kappa^{1/2}_{\Lambda_k}$.

Equation (31) is exactly the normalized pairing formula of proposition 1.45 applied to the representable source $\tau^{f,y}_{\Lambda_k}$.

If quotient descent also holds, then corollary 1.13 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}([\![\Xi^{(1)}_{\Lambda_k}]\!])$$

for every representative $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, which yields (32) after combining with (30). □

Theorem 1.50 (Fixed-cutoff seam witness criterion for descendant overlap)**.** *Fix an accepted tick k. Assume:*

1. *a detector-factorized witness-indexed seam realization at* Λ_k*;*

2. *a witness* y *with*
$$\overline{\mathrm{obs}}(y) \neq 0;$$

3. *the seam coefficient is nonzero,*
$$\widetilde{c}_{\Lambda_k} \neq 0;$$

4. *for the chosen filler tag* f*, the source* $\tau^{f,y}_{\Lambda_k}$ *is representable on the characterization layer in the sense of definition 1.43;*

5. *there exists a witness-indexed source-compatible descendant realization in the sense of definition 1.48.*

Then:

1. *the compressed welded LoG line is transversely physical at* Λ_k*, hence dominant-line realization holds at that cutoff;*

2. *the chosen 2T representative satisfies*
$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) \neq 0;$$

3. *therefore the transported descendant class has nonzero overlap with the canonically selected compressed dominant odd line at* Λ_k*;*

4. *if quotient descent also holds at* Λ_k*, then theorem 1.21 applies and the transported descendant class is dominant-active with well-defined sign* ε_{Λ_k}*.*

Proof. By theorem 1.41,
$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\widetilde{c}_{\Lambda_k}\,\overline{\mathrm{obs}}(y) \neq 0.$$
The same theorem therefore gives item (1): the compressed line is transversely physical, hence dominant-line realization holds at Λ_k.

By proposition 1.45,
$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$
Applying the comparison identity (26) now gives
$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \neq 0.$$
Because dominant-line realization has already been established, $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is nonzero and $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is defined. By proposition 1.18, this is equivalent to $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) \neq 0$, proving item (2).

Item (3) is exactly the nonzero-overlap condition from proposition 1.20. If quotient descent is also available, apply theorem 1.21 to obtain item (4). □

Smooth-taper branch continuity and witness transport

The previous fixed-cutoff theorem isolates the nonzero witness needed for activity. To transport that witness along an accepted branch one needs a continuity statement for the welded response. At theorem level, the clean analytic choice is the smooth receiver taper. The hard projector may still be used, but only on branch segments where its rank is constant.

Lemma 1.51 (Smooth-taper continuity of the receiver band projector)**.** *Let I be a connected accepted-branch segment. Suppose the receiver band is implemented by a smooth taper*

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda) = \sum_{\ell,m} w_\ell(\Lambda)\, |Y_{\ell m}\rangle\langle Y_{\ell m}|, \qquad 0 \le w_\ell(\Lambda) \le 1,$$

on the receiver one-particle space, where

$$\sup_{\ell\ge 0} |w_\ell(\Lambda') - w_\ell(\Lambda)| \longrightarrow 0 \qquad \text{as } \Lambda' \to \Lambda$$

for every $\Lambda \in I$. Then

$$\Lambda \longmapsto P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is continuous on I in operator norm. If instead the hard projector $\sum_{\ell\le\ell_\star(\Lambda)} |Y_{\ell m}\rangle\langle Y_{\ell m}|$ is used, the same conclusion holds on every connected subsegment of I on which $\ell_\star(\Lambda)$ is constant.

Proof. In the spherical-harmonic basis $\{Y_{\ell m}\}$, the difference

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is diagonal with eigenvalues $w_\ell(\Lambda') - w_\ell(\Lambda)$. Therefore

$$\left\| P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda) \right\|_{\mathrm{op}} = \sup_{\ell\ge 0} |w_\ell(\Lambda') - w_\ell(\Lambda)|.$$

The assumed uniform continuity of the taper coefficients is exactly the statement that this operator norm tends to zero as $\Lambda' \to \Lambda$.

For the hard projector, the coefficients take only the values 0 and 1, so discontinuities can occur only when $\ell_\star(\Lambda)$ jumps. On a subsegment where $\ell_\star(\Lambda)$ is constant, the projector is constant and therefore continuous. □

Theorem 1.52 (Smooth-taper branch continuity of the welded physical response)**.** *Let I be a connected accepted-branch segment. Write the welded source-to-covariance map in the UOM factorized form*

$$\mathsf{T}_\Lambda = \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda} \circ \mathsf{K}_{\mathrm{comp},\Lambda} \circ \mathsf{G}_{\mathrm{ret},\Lambda} \circ \mathsf{W}_{\mathrm{src},\Lambda}. \tag{33}$$

Assume:

1. *the receiver band is implemented by a smooth taper satisfying lemma 1.51;*
2. *the operator families*

$$\Lambda \longmapsto \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}, \qquad \Lambda \longmapsto \mathsf{K}_{\mathrm{comp},\Lambda}, \qquad \Lambda \longmapsto \mathsf{G}_{\mathrm{ret},\Lambda}, \qquad \Lambda \longmapsto \mathsf{W}_{\mathrm{src},\Lambda}, \qquad \Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}$$

   are continuous on I in operator norm on the corresponding band-limited source and covariance tangent spaces;

3. *the source family*

$$\Lambda \longmapsto \tau_\Lambda$$

is continuous on I in the welded source norm.

Then the physical covariance response

$$X_\Lambda[\tau_\Lambda] := \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda[\tau_\Lambda] \tag{34}$$

depends continuously on $\Lambda \in I$ in the Fisher norm. If the hard receiver projector is used instead, the same conclusion holds on each connected subsegment where $\ell_\star(\Lambda)$ is constant.

Proof. By lemma 1.51, the first factor $\Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}$ is operator-norm continuous along the branch. By assumption, the remaining three UOM factors in (33) and the physical projector $\Pi_{\mathrm{phys},\Lambda}$ are also operator-norm continuous. Hence the composed operator family

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda$$

is continuous in operator norm on I. Applying this continuous family to the continuous source vector τ_Λ gives continuity of $X_\Lambda[\tau_\Lambda]$ in the Fisher norm.

For the hard projector, the same argument applies on every connected subsegment where $\ell_\star(\Lambda)$ is constant, because there the first factor is constant. □

Corollary 1.53 (Branchwise continuity of the seam coefficient)**.** *Let I be a connected accepted-branch segment and assume:*

1. *the hypotheses of theorem 1.8 on I;*
2. *a detector-factorized branchwise seam realization on I with continuous unit source*

$$\Lambda \longmapsto \tau_\Lambda^{\mathrm{unit}};$$

3. *the hypotheses of theorem 1.52 for that unit source.*

Define the branchwise normalized seam coefficient by

$$c_\Lambda := \left\langle\!\!\left\langle \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda\big[\tau_\Lambda^{\mathrm{unit}}\big],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \right\rangle\!\!\right\rangle_{C_{*,\Lambda}}. \tag{35}$$

Then c_Λ is continuous on I. Consequently, for every fixed filler tag f and witness y, the normalized seam scalar

$$s_\Lambda(f, y) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

is continuous on I.

Proof. By theorem 1.52, the response vector

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda\big[\tau_\Lambda^{\mathrm{unit}}\big]$$

is continuous in the Fisher norm. By theorem 1.8, the normalized dominant vector $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$ is also continuous on I. Pairing these two continuous vectors in the Fisher metric gives continuity of c_Λ. The final statement is then immediate from (20). □

Theorem 1.54 (Branchwise seam witness transport)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of corollary 1.53;*

2. *quotient descent holds for every $\Lambda \in I$;*

3. *there exists a fixed LS witness y with*

$$\overline{\mathrm{obs}}(y) \neq 0;$$

4. *for some fixed filler tag f, there is a branchwise family of representatives*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that, for every $\Lambda \in I$, the corresponding seam source $\tau_\Lambda^{f,y}$ is representable on the characterization layer and the 2T-side witness-to-descendant comparison identity (27) holds.

Then:

1. *the branchwise dominant-response scalar of the chosen descendant representatives is*

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y); \tag{36}$$

2. *the scalar function $\Lambda \mapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})$ is continuous on I;*

3. *on every connected subsegment $I' \subset I$ on which $c_\Lambda \neq 0$, the sign*

$$\mathrm{sgn}\big(\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})\big)$$

is constant;

4. *because quotient descent holds, the same constant sign is the transported orientation sign of the descendant affine class on I'.*

Proof. For each $\Lambda \in I$, representability of $\tau_\Lambda^{f,y}$ and proposition 1.45 give

$$\rho_\Lambda^{\mathrm{dom}}\big(F_{\tau_\Lambda^{f,y},\Lambda}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda\big[\tau_\Lambda^{f,y}\big],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \big\rangle\!\big\rangle_{C_{*,\Lambda}} = s_\Lambda(f,y).$$

Applying the branchwise comparison identity (27) then yields

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = s_\Lambda(f,y).$$

Now use corollary 1.53 to substitute $s_\Lambda(f,y) = \sigma_{\mathrm{fill}}(f) c_\Lambda \overline{\mathrm{obs}}(y)$, which proves item (1). Item (2) follows because c_Λ is continuous and $\overline{\mathrm{obs}}(y)$ is a fixed nonzero scalar. On any connected subsegment where $c_\Lambda \neq 0$, the right-hand side of (36) is a continuous nonvanishing real-valued function, so its sign is constant. This proves item (3).

Finally, quotient descent implies that $\rho_\Lambda^{\mathrm{dom}}$ is constant on $[\Xi_\Lambda^{(1)}]$, so the sign of one representative is the sign of the whole descendant class. This gives item (4). □

Corollary 1.55 (Seam package discharge of the minimal scalar transport input)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of theorem 1.54. Then the branchwise family*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

satisfies

$$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

continuously on I. On every connected subsegment $I' \subset I$ such that

$$c_\Lambda \neq 0 \qquad \forall \Lambda \in I',$$

assumption (3) *of theorem 1.56 holds in its minimal scalar form with*

$$J_\Lambda^{\mathrm{desc}} := J_\Lambda^{f,y} \qquad (\Lambda \in I').$$

If, in addition, the three generator-wise vanishing statements of proposition 1.15 hold for every $\Lambda \in I'$, then theorem 1.56 applies on I'.

Proof. Continuity of the displayed scalar function is exactly item (2) of theorem 1.54, together with the explicit formula of item (1). On I', the hypotheses $c_\Lambda \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$ imply

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) \neq 0 \qquad \forall \Lambda \in I'.$$

Together with the continuity statement already proved above, this is exactly assumption (3) of theorem 1.56 in its minimal scalar form. The hypotheses of theorem 1.54 already include those of corollary 1.53, and hence the dominant-line persistence input of theorem 1.56. If the three generator-wise vanishing statements also hold on I', then every hypothesis of theorem 1.56 is in force there. □

Theorem 1.56 (Finite-cutoff Active Response Transport Theorem)**.** *Let I be a connected accepted-branch segment. For each $\Lambda \in I$, let*

$$[\Xi_\Lambda^{(1)}] := J_{0,\Lambda} + \mathcal{A}_\Lambda^- \subset E_\Lambda^-$$

be the parity-transported odd descendant affine class modulo the ambiguity space of definition 1.2. Assume:

1. *the hypotheses of theorem 1.8 hold on I;*
2. *for every $\Lambda \in I$, the three vanishing statements of proposition 1.15 hold;*
3. *there exists a branchwise family of representatives*

   $$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

   such that

   $$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) \neq 0 \qquad \forall \Lambda \in I,$$

   and such that the dominant-response scalar

   $$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}})$$

   is continuous on I.

Then, for every $\Lambda \in I$:

1. *the compressed dominant covariance tangent is nonzero,*

$$\delta C_\Lambda^{\mathrm{LoG}} \neq 0,$$

so the dominant line L_Λ^{dom} *of definition 1.3 is canonically defined;*

2. *the dominant-response coefficient* $\rho_\Lambda^{\mathrm{dom}}$ *annihilates the odd ambiguity space,*

$$\rho_\Lambda^{\mathrm{dom}}(A) = 0 \qquad \forall A \in \mathcal{A}_\Lambda^-;$$

3. *the transported descendant affine class carries a unique nonzero dominant-response value*

$$\rho_\Lambda^{\mathrm{dom}}(J) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big) \neq 0 \qquad \forall J \in [\Xi_\Lambda^{(1)}],$$

and hence a well-defined sign

$$\varepsilon_\Lambda := \mathrm{sgn}\Big(\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\Big) \in \{\pm 1\};$$

4. *the normalized orientation functional*

$$\chi_\Lambda := \nu_\Lambda^{-1}\rho_\Lambda^{\mathrm{dom}} \tag{37}$$

is well defined, annihilates $\mathcal{A}_\Lambda^-$*, and satisfies*

$$\chi_\Lambda(J) = \varepsilon_\Lambda \qquad \forall J \in [\Xi_\Lambda^{(1)}];$$

5. *along* I*, the sign* ε_Λ *is locally constant once an initial orientation is fixed.*

Proof. Item (1) is exactly the conclusion of theorem 1.8.

By assumption, the three generator-wise vanishing statements of proposition 1.15 hold for every $\Lambda \in I$. Therefore proposition 1.15 and theorem 1.16 give item (2) and the existence of the descended functional $\overline{\rho}_\Lambda^{\mathrm{dom}}$ at each cutoff.

Applying proposition 1.20 and theorem 1.21 pointwise to the nonvanishing scalar assumption $\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) \neq 0$ now yields item (3): the entire descendant affine class has one common nonzero dominant-response value, and its sign ε_Λ is well defined.

Since $\nu_\Lambda > 0$ by item (3), the normalized functional χ_Λ of (37) is well defined. Item (2) implies that χ_Λ annihilates $\mathcal{A}_\Lambda^-$, and item (3) gives

$$\chi_\Lambda(J) = \frac{\rho_\Lambda^{\mathrm{dom}}(J)}{\left|\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\right|} = \varepsilon_\Lambda \qquad \forall J \in [\Xi_\Lambda^{(1)}],$$

proving item (4).

Finally, assumption (3) provides a continuous nonvanishing real-valued branchwise descendant scalar, while items (1)–(3) above identify that scalar with the transported orientation sign of the full descendant affine class. Since $\Lambda \mapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}})$ is continuous and nonzero on the connected set I, its sign is constant, hence in particular locally constant. Using item (3), this is exactly the transported orientation sign of the full descendant affine class, proving item (5). □

Proposition 1.57 (Ambiguity-triviality on the compressed welded image)**.** *Assume the hypotheses of theorem 1.56 at an accepted tick k. Define the reduced welded image of an odd generator $F \in E^-_{\Lambda_k}$ by*

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) := \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k},$$

where $P^{\mathrm{LoG}}_{-,\Lambda_k}$ is the canonical compressed odd projector of **??**. *Then*

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) \qquad \forall F \in E^-_{\Lambda_k},\ \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Equivalently, $\mathsf{W}^{\mathrm{red}}_{\Lambda_k}$ factors through the ambiguity quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$. In particular, if $J_1, J_2 \in [\Xi^{(1)}_{\Lambda_k}]$, then

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_1) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_2).$$

Proof. Item (2) of theorem 1.56 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = 0 \qquad \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Therefore, for every $F \in E^-_{\Lambda_k}$ and $A \in \mathcal{A}^-_{\Lambda_k}$,

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F + A)\, P^{\mathrm{LoG}}_{-,\Lambda_k} = \big(\rho^{\mathrm{dom}}_{\Lambda_k}(F) + \rho^{\mathrm{dom}}_{\Lambda_k}(A)\big) P^{\mathrm{LoG}}_{-,\Lambda_k} = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k}.$$

Since **??** fixes $P^{\mathrm{LoG}}_{-,\Lambda_k}$ canonically on the compressed, co-centered, band-locked welded channel, the right-hand side depends only on the ambiguity class of F. If $J_2 - J_1 \in \mathcal{A}^-_{\Lambda_k}$, apply the same identity with $F = J_1$ and $A = J_2 - J_1$. □

Proposition 1.58 (Construction of an admissible calibration for the minimal transported interface)**.** *Fix an accepted tick k. Suppose a weak transported-interface package is given:*

$$\mathcal{A}^-_{\Lambda_k} \subset E^-_{\Lambda_k}, \qquad [\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k},$$

together with a continuous nonzero linear functional

$$\chi_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

and a sign $\varepsilon_{\Lambda_k} \in \{\pm 1\}$ such that

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \chi_{\Lambda_k}, \qquad \chi_{\Lambda_k}(J) = \varepsilon_{\Lambda_k} \quad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

Assume in addition that $\ker \chi_{\Lambda_k}$ admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it. Choose any vector $J^\sharp_{\Lambda_k} \in E^-_{\Lambda_k}$ satisfying

$$\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1,$$

any scale $\beta_{\Lambda_k} > 0$, and any continuous symmetric coercive bilinear form

$$\langle \cdot, \cdot \rangle^{\mathrm{aux},\chi}_{\Lambda_k} : \ker \chi_{\Lambda_k} \times \ker \chi_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\ker \chi_{\Lambda_k}$. Define

$$\mathsf{B}_{\Lambda_k}(F, G) := \beta_{\Lambda_k}\, \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \Big\langle F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k},\, G - \chi_{\Lambda_k}(G) J^\sharp_{\Lambda_k} \Big\rangle^{\mathrm{aux},\chi}_{\Lambda_k}. \tag{38}$$

Then B_{Λ_k} *is a continuous symmetric positive definite bilinear form on* $E^-_{\Lambda_k}$ *whose induced norm Hilbertizes* $E^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0 \qquad \forall K \in \ker \chi_{\Lambda_k}.$$

Hence $\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0$ *in particular for every* $K \in \mathcal{A}^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \beta_{\Lambda_k}.$$

Moreover, for any representative $J^\Xi_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$*, the choice*

$$J^\sharp_{\Lambda_k} := \varepsilon_{\Lambda_k} J^\Xi_{\Lambda_k}$$

is admissible.

Proof. Because $\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1$, the linear map

$$P^\chi_{\Lambda_k}(F) := F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k}$$

takes values in $\ker \chi_{\Lambda_k}$:

$$\chi_{\Lambda_k}\big(P^\chi_{\Lambda_k}(F)\big) = \chi_{\Lambda_k}(F) - \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 0.$$

Thus every $F \in E^-_{\Lambda_k}$ has the decomposition

$$F = \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k} + P^\chi_{\Lambda_k}(F), \qquad P^\chi_{\Lambda_k}(F) \in \ker \chi_{\Lambda_k}.$$

This decomposition is unique: if $F = aJ^\sharp_{\Lambda_k} + K$ with $K \in \ker \chi_{\Lambda_k}$, applying χ_{Λ_k} gives $a = \chi_{\Lambda_k}(F)$.

Equation (38) is therefore well defined, continuous, and symmetric. If $\mathsf{B}_{\Lambda_k}(F, F) = 0$, then

$$\beta_{\Lambda_k} \chi_{\Lambda_k}(F)^2 = 0$$

and

$$\Big\langle P^\chi_{\Lambda_k}(F), P^\chi_{\Lambda_k}(F) \Big\rangle^{\mathrm{aux},\chi}_{\Lambda_k} = 0.$$

Since $\beta_{\Lambda_k} > 0$, we get $\chi_{\Lambda_k}(F) = 0$, hence $P^\chi_{\Lambda_k}(F) = F$. Coercivity on $\ker \chi_{\Lambda_k}$ then implies $F = 0$, proving positive definiteness.

The map

$$F \longmapsto \big(\chi_{\Lambda_k}(F), P^\chi_{\Lambda_k}(F)\big)$$

is a continuous linear isomorphism from $E^-_{\Lambda_k}$ onto $\mathbb{R} \oplus \ker \chi_{\Lambda_k}$, with inverse

$$(a, K) \longmapsto aJ^\sharp_{\Lambda_k} + K.$$

Under this isomorphism, B_{Λ_k} becomes the product Hilbert form

$$(a, K), (b, L) \longmapsto \beta_{\Lambda_k} ab + \langle K, L\rangle^{\mathrm{aux},\chi}_{\Lambda_k},$$

so its induced norm Hilbertizes $E^-_{\Lambda_k}$.

Finally, if $K \in \ker \chi_{\Lambda_k}$, then $P^\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 0$ and $\chi_{\Lambda_k}(K) = 0$, hence

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0.$$

Setting $K = J^\sharp_{\Lambda_k}$ gives $\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \beta_{\Lambda_k}$. If $J^\Xi_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, then $\chi_{\Lambda_k}(J^\Xi_{\Lambda_k}) = \varepsilon_{\Lambda_k}$, so $J^\sharp_{\Lambda_k} := \varepsilon_{\Lambda_k} J^\Xi_{\Lambda_k}$ satisfies $\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1$. □

Remark 1.59 (Calibration existence versus stiffness matching). Proposition 1.58 is a construction theorem, not an unconditional existence theorem: it extends a previously chosen Hilbertizing form on $\ker \chi_{\Lambda_k}$ to an admissible bilinear form on the whole weak transported interface. It does *not* identify the scale β_{Λ_k} with the split/leak stiffness $\widetilde{\alpha}(\Lambda_k)$. That physical identification belongs to a later strong or physical layer.

Theorem 1.60 (Rank-one descendant-sector upgrade)**.** *Assume theorem 1.56, and fix an accepted tick k on the branch. Suppose in addition that on the transported descendant sector the projected response map*

$$F \longmapsto \Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F), \qquad \Pi^{\mathrm{dom}}_{\Lambda_k} := \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \otimes \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k},$$

has kernel exactly $\mathcal{A}^-_{\Lambda_k}$. Then the stronger conclusions follow:

1. *the descendant quotient is rank one,*

$$E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k} \cong \mathbb{R};$$

2. *the orientation kernel is exactly the ambiguity space,*

$$\ker \chi_{\Lambda_k} = \ker \rho^{\mathrm{dom}}_{\Lambda_k} = \mathcal{A}^-_{\Lambda_k},$$

   so the descendant sector satisfies the rank-one kernel-identification package with quotient generated by any vector $J^{\mathrm{act}}_{\Lambda_k}$ satisfying $\chi_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$;

3. *after choosing any such $J^{\mathrm{act}}_{\Lambda_k}$, any positive scale β_{Λ_k}, and any auxiliary coercive Hilbertizing form on $\ker \chi_{\Lambda_k}$, proposition 1.58 yields an admissible calibration form. Recovering the stiffness-matched formula of **??** requires additional physical identifications, namely $\chi_{\Lambda_k} = \rho_{\Lambda_k}$, $\ker \chi_{\Lambda_k} = \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$, agreement of the chosen normalized generator with the split/leak active generator, and $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$.*

Proof. By theorem 1.56, the dominant line is defined at Λ_k, the scalar ν_{Λ_k} is positive, and

$$\chi_{\Lambda_k} = \nu^{-1}_{\Lambda_k} \rho^{\mathrm{dom}}_{\Lambda_k}.$$

Hence $\ker \chi_{\Lambda_k} = \ker \rho^{\mathrm{dom}}_{\Lambda_k}$.

For every $F \in E^-_{\Lambda_k}$, the definition of $\rho^{\mathrm{dom}}_{\Lambda_k}$ gives

$$\Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}.$$

Therefore the kernel of the projected response map is exactly $\ker \rho^{\mathrm{dom}}_{\Lambda_k}$. By the additional hypothesis, that kernel is $\mathcal{A}^-_{\Lambda_k}$, proving item (2).

By theorem 1.56, there exists $J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ with $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0$, hence the projected response map is nonzero. Its image is automatically contained in the one-dimensional line $L^{\mathrm{dom}}_{\Lambda_k}$, so the first isomorphism theorem gives

$$E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k} \cong \mathrm{im}\big(\Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}\big) \cong \mathbb{R},$$

which is item (1). Any vector $J^{\mathrm{act}}_{\Lambda_k}$ with $\chi_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$ lies outside $\ker \chi_{\Lambda_k} = \mathcal{A}^-_{\Lambda_k}$, so its quotient class generates this one-dimensional quotient, completing item (2).

Item (3) is now an application of proposition 1.58 with the weak-interface data supplied by theorem 1.56 and the chosen normalized vector $J^{\mathrm{act}}_{\Lambda_k}$. The final sentence identifies when this weak calibration construction coincides with the stiffness-matched formula of **??**. □